Supporting Information: A First-Principles Investigation of Janus HfSSe and Co-Doped HfSSe Monolayers as Anode Materials for Lithium- and Sodium-Ion Batteries

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S1. Plane-Wave Energy Cutoff and k-Point Convergence Tests

In plane-wave density functional theory (DFT) calculations, the accuracy of the calculated total energy depends strongly on the plane-wave kinetic-energy cutoff and the density of the reciprocal-space sampling. The plane-wave kinetic-energy cutoff, E_{cut}, corresponding to `ecutwfc` in Quantum ESPRESSO, defines the maximum kinetic energy of the plane waves included in the basis set used to expand the Kohn–Sham wave functions. Increasing E_{cut} generally improves the completeness of the plane-wave basis and hence the numerical accuracy of the calculation, although at the expense of greater computational cost. Similarly, the k-point mesh determines the sampling density of the Brillouin zone. A sufficiently dense k-point grid is required to obtain converged electronic energies and related material properties while avoiding unnecessary computational expense.

To establish suitable computational parameters for the subsequent DFT calculations, systematic convergence tests were performed with respect to both E_{cut} and the k-point sampling density. The corresponding total energies are presented in Figure S1.

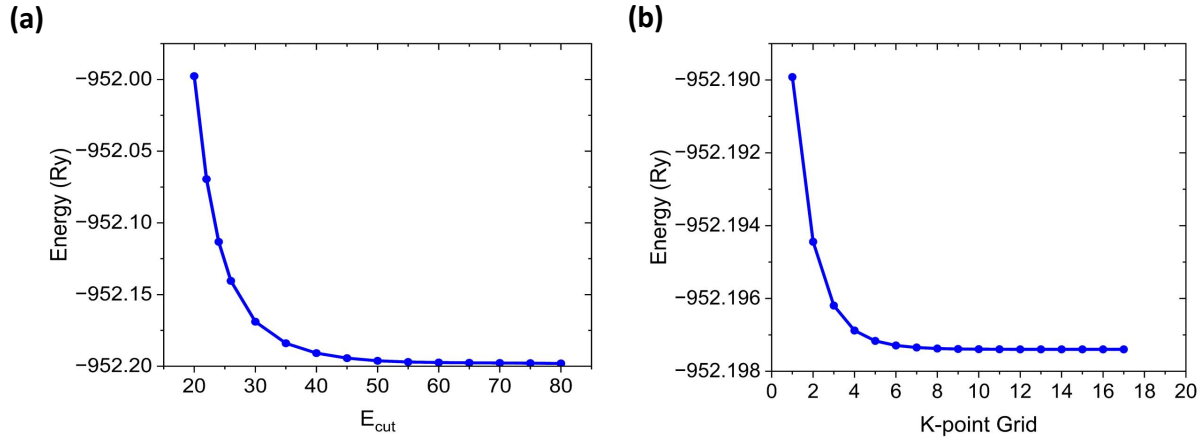


Figure S1: Convergence of the DFT total energy with respect to (a) the plane-wave kinetic-energy cutoff, E_{cut}, and (b) the k-point sampling density. For the k-point convergence test, the horizontal-axis value N represents an $N \times N \times 1$ Monkhorst–Pack mesh.

As shown in Figure S1(a), the calculated total energy decreases substantially when E_{cut} is increased from 20 Ry, reflecting the progressive improvement in the completeness of the plane-wave basis. The magnitude of the energy variation decreases rapidly with increasing cutoff energy, and

the total energy approaches a nearly constant value at higher cutoff energies. In particular, beyond approximately $E_{\mathrm{cut}} = 60$ Ry, further increases in the cutoff produce only negligible changes in the calculated total energy. Therefore, $E_{\mathrm{cut}} = 60$ Ry was selected as an appropriate compromise between numerical accuracy and computational efficiency for the subsequent calculations

S2. Projected Density of States after Single-Li Incorporation

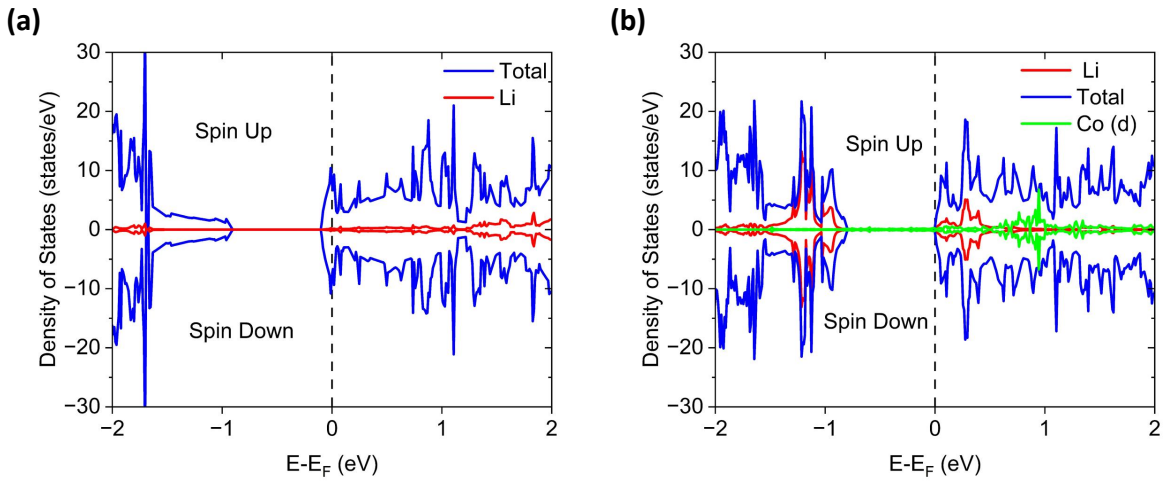


Figure S2: Spin-polarized density of states after incorporation of a single Li atom into (a) the pristine monolayer and (b) the Co-doped monolayer.

Spin-polarized density of states (DOS) calculations were performed after introducing a single Li atom into the pristine and Co-doped monolayers. The resulting total and projected density of states (PDOS) are presented in Figure S2. The Fermi energy (E_{F}) was shifted to zero energy for comparison, while the positive and negative DOS components represent the spin-up and spin-down channels, respectively. As shown in Figure S2(a), Li incorporation modifies the electronic states of the pristine monolayer, with the Li-projected states contributing primarily away from the Fermi level. For the Co-doped monolayer [Figure S2(b)], additional electronic states associated with the Co d orbitals are observed, particularly in the vicinity of the Fermi level. The differences between the spin-up and spin-down DOS further indicate the spin-dependent nature of the electronic structure.

S3. Hf-to-Hf Migration Pathway on Pristine HfSSe

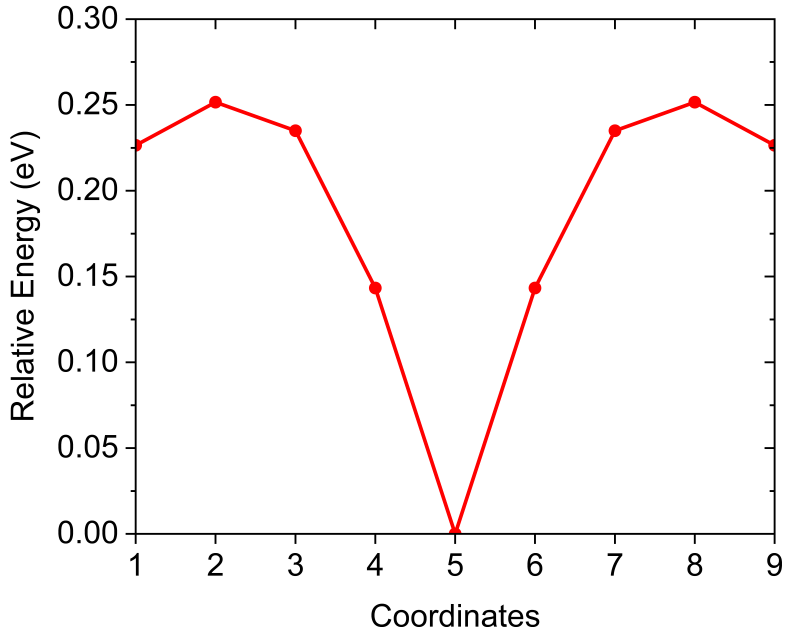


Figure S3: Calculated relative-energy profile along the Hf-to-Hf migration pathway on the pristine HfSSe monolayer.

The migration pathway between two neighboring Hf sites on the pristine HfSSe monolayer was investigated by evaluating the relative energy along the corresponding diffusion path. The calculated energy profile is presented in Figure S3. As shown in Figure S3, the relative energy varies systematically along the Hf-to-Hf migration pathway, with the lowest-energy configuration occurring near the center of the path. The maximum energy difference along the pathway is approximately 0.25 eV, corresponding to a migration barrier of about 0.25 eV. This relatively low energy barrier indicates an energetically accessible migration pathway across the pristine HfSSe surface.